CSCI 164 Project

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DATASETS USED IN PROJECT

One for Classification Models and another for Regression models

DATASET 1: IMDb Top 1000 (Regression)

- Features
  - IMDb ratings, Directors, No_of_votes...
- Initially
  - Tried to Predict IMDb ratings
  - Model could not get good results
  - A Lot of trouble...
- Afterwards
  - Changed Goal: predicting Gross

DATASET 2: Titanic (Classification)

- Features
  - Class, Gender, Fare...
- Simple in comparison to the IMDb dataset
-
- Goal of Predicting survivors of the titanic

IMDb Top 1000: Preprocess

- Rather difficult Dataset
  - Not many numerical features
    - IMDB_RATING, META_SCORE, GROSS, RUNTIME, NO_OF_VOTES, RELEASED_YEAR
  - Not the strongest of Datasets for this project
- Preprocessing
  - Many rows had missing values such as Gross or Runtimes
    - Median Imputation 😨
  - Data stored as objects and not numeric
    - Runtime and Gross
  - Directors and Genres
    - Encoded and split into different columns
    - Only top Directors were added...
  - Log Gross
    - Using another work I was able to improve my results drastically by applying log to Gross. Making it able to learn more from it

Before Preprocessing

```
Data columns (total 15 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Poster_Link    1000 non-null   object
 1   Series_Title   1000 non-null   object
 2   Released_Year  1000 non-null   object
 3   Certificate    899 non-null    object
 4   Genre          1000 non-null   object
 5   Runtime        1000 non-null   object
 6   IMDB_Rating    1000 non-null   float64
 7   Meta_score     843 non-null    float64
 8   Director       1000 non-null   object
 9   Star1          1000 non-null   object
 10  Star2          1000 non-null   object
 11  Star3          1000 non-null   object
 12  Star4          1000 non-null   object
 13  No_of_Votes    1000 non-null   int64
 14  Gross          831 non-null    object
dtypes: float64(2), int64(1), object(12)
memory usage: 117.3+ KB
```

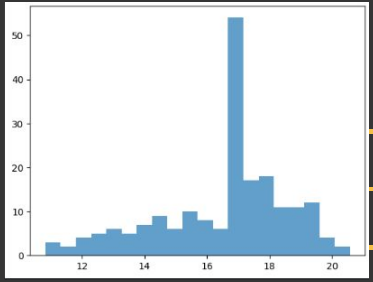
After Preprocessing

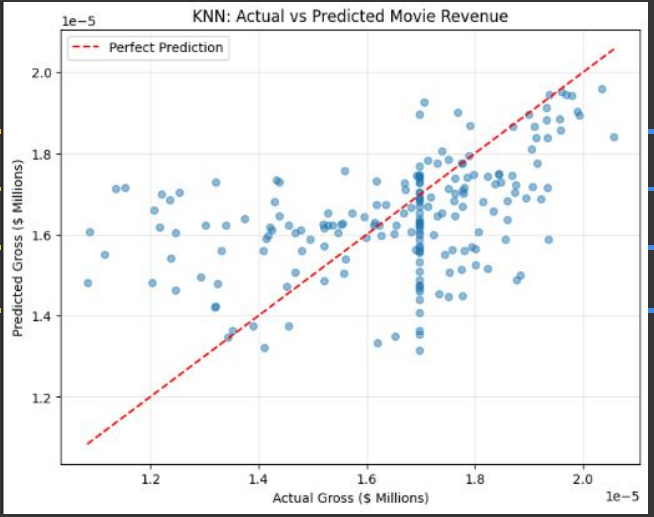
```
 0   Released_Year                  999 non-null   float64
 1   Runtime                        999 non-null   int64
 2   IMDB_Rating                    999 non-null   float64
 3   Meta_score                     999 non-null   float64
 4   Director                       999 non-null   object
 5   Star1                          999 non-null   object
 6   Star2                          999 non-null   object
 7   Star3                          999 non-null   object
 8   Star4                          999 non-null   object
 9   No_of_Votes                    999 non-null   int64
 10  Gross                          999 non-null   float64
```

...

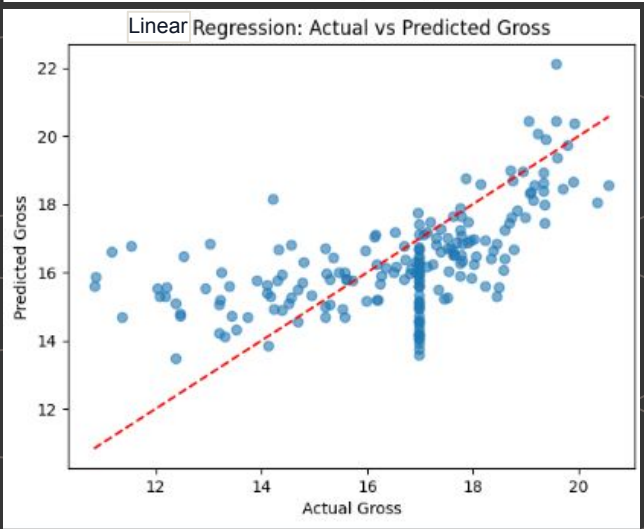
```
 32  Top_Director_Akira Kurosawa      999 non-null   bool
 33  Top_Director_Alfred Hitchcock    999 non-null   bool
 34  Top_Director_Billy Wilder        999 non-null   bool
 35  Top_Director_Christopher Nolan   999 non-null   bool
 36  Top_Director_Clint Eastwood      999 non-null   bool
 37  Top_Director_Hayao Miyazaki      999 non-null   bool
 38  Top_Director_Martin Scorsese     999 non-null   bool
 39  Top_Director_Other               999 non-null   bool
 40  Top_Director_Stanley Kubrick     999 non-null   bool
 41  Top_Director_Steven Spielberg    999 non-null   bool
 42  Top_Director_Woody Allen         999 non-null   bool
dtypes: bool(32), float64(4), int64(2), object(5)
```

IMDb Top 1000: Models/Results





- Models used
  - KNN and Linear Regression
    - Linear Regression was typically better
      - Features had relatively linear relationships
        - High Ratings -> High Gross
    - KNN
      - Many outliers and not best dataset for it due to similarity not necessarily meaning high gross
- Results
  - Better than what I was expecting...
  - Below Average to Poor performance
    - Difficult to predict as there are many factors outside of IMDb dataset that affect the gross
    - Dataset is inherently skewed
      - Top 1000 movies...
      - Low rating movies minimal
    - Median Imputation 😨



Titanic Dataset: Preprocess

- Luckily very clean dataset unlike IMDb
  - Very minimal cleaning needed
    - Removal of unnecessary features
      - Unnamed and PassengerId

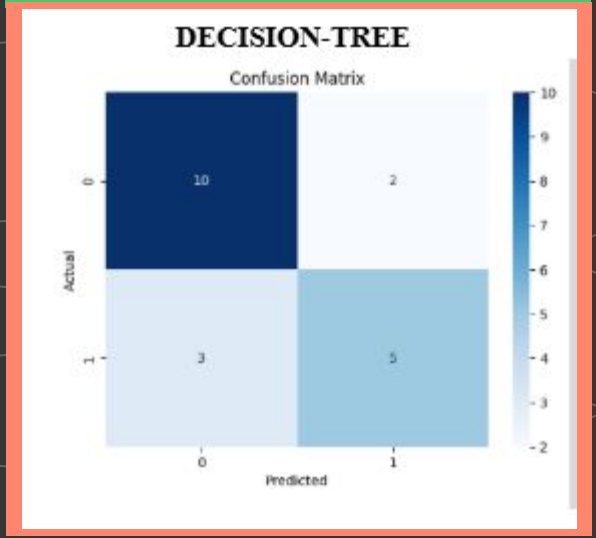
```
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Unnamed: 0   100 non-null    int64
 1   PassengerId  100 non-null    int64
 2   Survived     100 non-null    int64
 3   Sex          100 non-null    int64
 4   Age          100 non-null    float64
 5   Fare         100 non-null    float64
 6   Pclass_1     100 non-null    int64
 7   Pclass_2     100 non-null    int64
 8   Pclass_3     100 non-null    int64
 9   Family_size  100 non-null    float64
 10  Title_1      100 non-null    int64
 11  Title_2      100 non-null    int64
 12  Title_3      100 non-null    int64
 13  Title_4      100 non-null    int64
 14  Emb_1        100 non-null    int64
 15  Emb_2        100 non-null    int64
 16  Emb_3        100 non-null    int64
dtypes: float64(3), int64(14)
memory usage: 13.4 KB
```

	Survived	Sex	Age	Fare	Pclass_1	Pclass_2	Pclass_3	Family_size	Title_1	Title_2	Title_3
0	0	1	0.2000	0.050749	0	1	0	0.0	1	0	0
1	0	0	0.3500	0.135753	0	0	1	1.0	0	0	0
2	0	1	0.3500	0.059914	1	0	0	0.0	1	0	0
3	0	1	0.3125	0.015412	0	0	1	0.0	1	0	0
4	0	1	0.4875	0.025374	0	1	0	0.0	1	0	0

Not much to fix up luckily!

Titanic Dataset: Models/Results

- Models used
  - Logistic Regression and Decision Tree
    - Overall both performed well with Decision Tree performing slightly worse due to limited dataset size
    - Decision Tree Accuracy: **0.75**
    - Logistic Regression Accuracy: **0.95**
  - Issues with dataset size, only 100 inputs
    - 80 for training and 20 for testing
  - Puts decision tree at disadvantage
    - Lacks proper size to develop patterns
- Attempted to improve results of Decision tree using grid Tuning.



Sadly no improvements 😿

```
Accuracy: 0.75
              precision    recall  f1-score   support

           0       0.77      0.83      0.80        12
           1       0.71      0.62      0.67         8

    accuracy                           0.75        20
   macro avg       0.74      0.73      0.73        20
weighted avg       0.75      0.75      0.75        20
```

```
param_grid = {
    "max_depth": [3, 5, 10],
    "min_samples_split": [2, 5, 10]
}

grid = GridSearchCV(DecisionTreeClassifier(random_state=42),
                    param_grid,
                    cv=5)

grid.fit(X_train, y_train)
```

Conclusions

- Dataset selection is important!
  - Understanding which datasets are effective requires both looking at the features and understanding real relationships between them
    - Titanic and IMDb Top 1000 showcase that well

- Media Imputation can be helpful in smaller amounts
  - Too many can cause issues in the results of a model

- Log(Gross) for the IMDb helped to reduce skew of high grossing films